RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Even Semester)

INNOVATIVE TEACHING METHOD

Degree, Semester& Branch: II Semester B.E. ECE B

Course Code & Title: EC8252 Electronic Devices

Name of the Faculty member: Mrs.S.Jeeva

Activity: Visual Quiz

Topic: Junction Field Effect Transistors

Reference: https://www.sanfoundry.com/electronic-devices-circuits-questions-answers-pn-junction-diode/

Justification:

The concept of Field Effect Transistor plays a vital role in forthcoming courses in the higher semesters. In order to assess the understanding level of the students, visible quiz activity was conducted.

Time Allotted for the Activity: 10 Minutes

Details of the Implementation:

- Initially, 10 questions covering the important topic in the unit was taken and prepared in a power point presentation.
- The students were split into a group of four members (class strength: 39). 10 groups were formed and each member of the group were given with a card containing A, B, C, D as shown below.

- Each group was provided with a paper to write down the answer for each question.
- Once the question is displayed one by one. The group member should raise the correct option card for the question.

VISUAL QUIZ

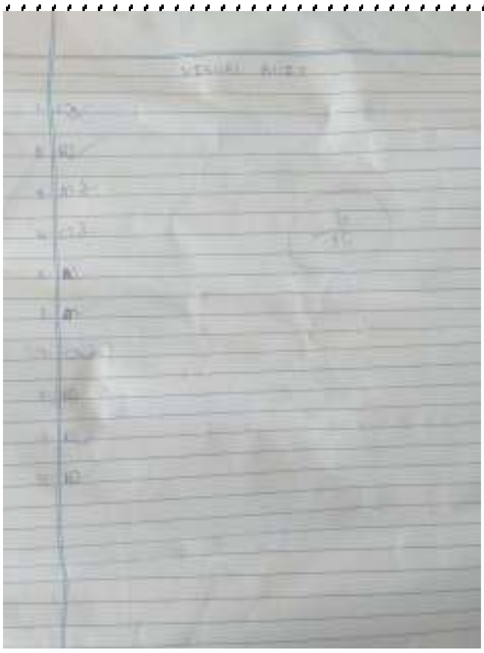
FIELD EFFECT TRANSISTOR

1. An FET has

A. Very High Input resistance
B. Very Low input resistance
C. High connection emitter junction
D. Forward Biased PN Junction

4.The pinch-off voltage for a N-channel JFET is 4V. When V_{GS} = 1 V, the at $|V_{DS(min)}|$ which the pinch-off occurs is equal to

A. 3 V
B. 5 V
C. 4 V
D. 1 V



- The other members will write the team answer in the paper given. Finally, the paper was interchanged with the group and finally discussed the correct answers.

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was measured by the performance in the internal Assessment Test - II
- ✓ After completing the activity, the corrected paper with mistakes highlighted were given to the teams and discussed the correct answers.

Reflective Critique:

Benefits of conducting this activity:

- This activity helped to quickly recap the important fundament concepts related to FET.
- Visible quiz is a team-based activity which supports collaborative learning through which the concepts will be made clear.

Challenges:

- This activity was planned for 10 minutes but it took 15 minutes for completing the discussion.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	1	1	-	-	-	-	2	-	3	-	2

CO3: The Students will be able to analyze the operation of various Field Effect Transistors.

References:

1. Salivahanan S, Suresh Kumar N , Vallavaraj A, ‘ Electronic devices and circuits, 3 rd edition Tata Mcgraw Hill, 2008.
2. https://www.sanfoundry.com/electronic-devices-circuits-questions-answers-pn-junction-diode/

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Degree, Semester& Branch: II Semester B.E. ECE B

Course Code & Title: EC8252 Electronic Devices

Name of the Faculty member: Mrs.S.Jeeva

Activity: Just a Minute Paper (JAM)

Topic: h-Parameter Model

Reference: Salivahanan S, Suresh Kumar N , Vallavaraj A, ‘ Electronic devices and circuits, 3 rd edition Tata Mcgraw Hill, 2008.

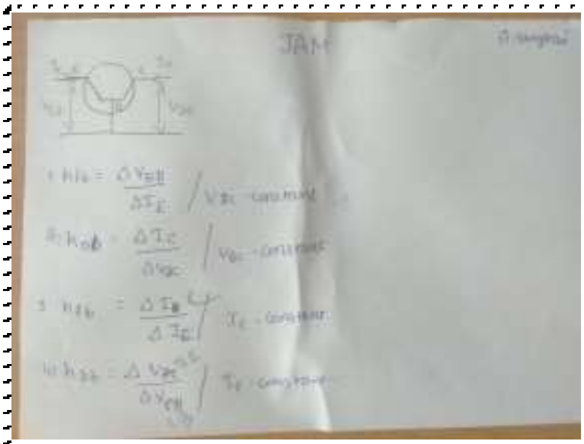
Justification:

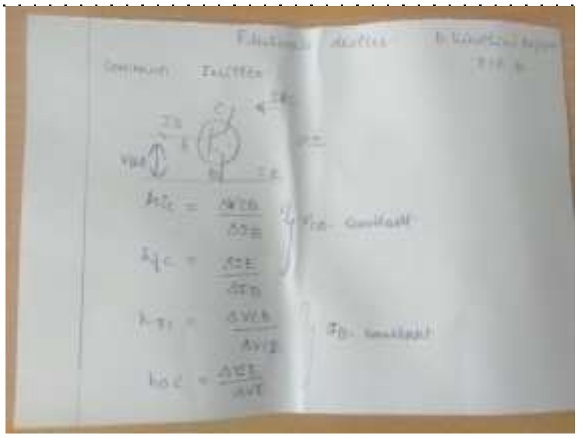
The concept of h parameter model is important and will be used in future courses like Electronic circuits I, Linear Integrated circuits. The common base current gain, common emitter current gain and relationship between alpha and beta will always be confused. Inorder to avoid such confusion this activity is conducted.

Time Allotted for the Activity: 5 Minutes

Details of the Implementation:

After teaching the concept of h parameter model, the students were given with a paper and asked to write the h-parameter for Common base configuration.





Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when most of the students wrote correctly.
- ✓ After completing the activity, the corrected paper with mistakes highlighted were given to the students.

Reflective Critique:

Benefits of conducting this activity:

- The concept of h parameter model can be easily applied for solving problems.
- It helps the student to perform modelling while designing amplifiers.

Challenges:

- Few students got confused. Once the paper has been evaluated and given to the students understood the mistake done. The time for discussing the concept in the next class is more.

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	1	-	-	-	-	2	-	2	-	2

CO2: The students will be able to describe the basic geometry, operation and various configuration of Bipolar Junction Transistor.

References:

1. La Rudi, 'Application of Teaching Model of Team Assisted Individualization (TAI) In Basic Chemistry Courses in Students of Forestry and Science of Environmental UniverstiasHalu Oleo', International Journal of Education and Research,Vol. 5 No. 11 November 2017, pp 69-76.
2. Salivahanan S, Suresh Kumar N , Vallavaraj A, ' Electronic devices and circuits, 3 rd edition Tata Mcgraw Hill, 2008.